

Qualification Pack

Assistant Computer Technician

QP Code: NIE/ELE/Q0904

Version: 1.0

NSQF Level: 3

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NIE/ELE/Q0904: Assistant Computer Technician

Brief Job Description

The Assistant Computer Technician certificate course is designed to equip learners with the essential skills for assembling, troubleshooting, and installing computer hardware, software, and peripheral devices. Developed in collaboration with industry experts, the course prepares students for both employment and entrepreneurial opportunities in the IT hardware field.

Personal Attributes

The Assistant Computer Technician course develops key attributes like technical expertise, attention to detail, and problem-solving abilities. Students learn to identify and configure hardware and software, install devices, follow safety procedures, and troubleshoot system issues. They also gain skills in managing operating systems, ensuring security through antivirus software, and performing diagnostic tasks with precision and analytical thinking.

Applicable National Occupational Standards (NOS)

Compulsory NOS:

1. [NIE/ELE/N0918: Identification of Computer Hardware Components and Peripherals devices](#)
2. [NIE/ELE/N0920: Installation of Computer Hardware Components and setting up of peripherals devices](#)
3. [NIE/ELE/N0919: Installation and Configuration of Computer Software](#)
4. [NIE/ELE/N0917: Troubleshooting and Maintenance of Computer System](#)
5. [NIE/ELE/N0921: Comprehensive Computer System Setup, Installation, and Maintenance Project](#)
6. [DGT/VSQ/N0101: Employability Skills \(30 Hours\)](#)

Qualification Pack (QP) Parameters

Sector	Electronics
Sub-Sector	Electronics Assembly and Services
Occupation	Electronics Assembly
Country	India

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NSQF Level	3
Credits	11
Aligned to NCO/ISCO/ISIC Code	3
Minimum Educational Qualification & Experience	10th Class (Pass) with NA of experience OR Grade 8 pass with 2 years of (NTC/ NAC) after 8th with NA of experience OR 8th grade pass with 3 Years of experience relevant experience OR Previous relevant Qualification of NSQF Level (2) with 1.5 years of experience relevant experience
Minimum Level of Education for Training in School	
Pre-Requisite License or Training	NA
Minimum Job Entry Age	16 Years
Last Reviewed On	NA
Next Review Date	25/07/2027
NSQC Approval Date	25/07/2024
Version	1.0
Reference code on NQR	QG-03-EH-02853-2024-V2-NIELIT
NQR Version	2

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NIE/ELE/N0918: Identification of Computer Hardware Components and Peripherals devices

Description

This module provides a comprehensive understanding of different types of computers, their working principles, and the various hardware components that form the core of a computer system. Students will learn to distinguish between hardware and software components, identify input/output devices, and recognize different types of motherboards, memory systems (RAM, ROM), storage devices, and computer ports and connectors. The module combines theoretical knowledge with practical skills, enabling students to install and configure hardware such as motherboards, memory, storage, and peripherals like printers, keyboards, and mice. Through hands-on experience with tools and equipment, students will be prepared to effectively manage, troubleshoot, and support computer hardware systems in real-world settings.

Scope

The scope covers the following :

- The NOS includes the identification and understanding of various computer types, hardware components, and peripheral devices. It covers differentiating between hardware and software, identifying input/output devices, and recognizing components such as motherboards, memory systems, storage devices, and computer ports. This NOS also focuses on developing practical skills for installing, configuring, and managing these components, ensuring students are equipped to handle real-world computer hardware tasks, including troubleshooting and maintenance.

Elements and Performance Criteria

Identification of Computer Types and Hardware Components

To be competent, the user/individual on the job must be able to:

- PC1.** Students shall identify different types of computers, including desktop, laptop, and server systems, and describe their working principles.
- PC2.** Students shall differentiate between hardware and software components, explaining the function of key hardware elements such as the CPU, motherboard, and RAM.

Recognition and Installation of Input/Output Devices and Peripherals

To be competent, the user/individual on the job must be able to:

- PC3.** Students shall recognize common input/output devices (e.g., keyboard, mouse, printer, monitor) and explain their roles in a computer system.
- PC4.** Students shall demonstrate the ability to install and configure various input/output devices and peripheral components.

Identification and Configuration of Motherboards, Memory, and Storage Devices

To be competent, the user/individual on the job must be able to:

- PC5.** Students shall identify different types of motherboards, memory (RAM, ROM), and storage devices (HDD, SSD) and explain their functions within the computer system.
- PC6.** Students shall demonstrate the installation and configuration of motherboards, memory modules, and storage devices.

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Identification and Use of Computer Ports and Connectors

To be competent, the user/individual on the job must be able to:

- PC7.** Students shall identify different types of computer ports (USB, HDMI, Ethernet) and connectors and explain their purposes.
- PC8.** Students shall demonstrate the proper connection of peripherals and devices using various ports and connectors, ensuring correct setup and functionality.

Knowledge and Understanding (KU)

The individual on the job needs to know and understand:

- KU1.** Understand the various types of computers (desktops, laptops, servers) and their specific functions, along with the fundamental principles of how they operate, integrating hardware and software.
- KU2.** Recognize the differences between hardware and software components, including key hardware elements like the CPU, motherboard, and RAM, and their roles in system performance.
- KU3.** Identify common input (keyboards, mice) and output devices (monitors, printers) and comprehend the processes involved in their installation and configuration for proper operation.
- KU4.** Gain knowledge of different motherboard types, various memory (RAM, ROM), and storage systems (HDD, SSD), as well as the various computer ports and connectors, understanding their purposes and specifications for effective connectivity.

Generic Skills (GS)

User/individual on the job needs to know how to:

- GS1.** Develop the ability to analyze and assess various types of computers and their components, enabling informed decisions regarding hardware selection and configuration.
- GS2.** Acquire hands-on skills in installing and configuring computer hardware, including motherboards, memory systems, and peripherals, enhancing technical competence in real-world scenarios.
- GS3.** Cultivate problem-solving abilities to troubleshoot and resolve issues related to hardware components and connectivity, ensuring efficient system performance.
- GS4.** Strengthen attention to detail when identifying components, connectors, and installation processes, promoting accuracy and thoroughness in technical tasks.

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Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
<i>Identification of Computer Types and Hardware Components</i>	50	22	-	-
PC1. Students shall identify different types of computers, including desktop, laptop, and server systems, and describe their working principles.	-	-	-	-
PC2. Students shall differentiate between hardware and software components, explaining the function of key hardware elements such as the CPU, motherboard, and RAM.	-	-	-	-
<i>Recognition and Installation of Input/Output Devices and Peripherals</i>	-	-	-	-
PC3. Students shall recognize common input/output devices (e.g., keyboard, mouse, printer, monitor) and explain their roles in a computer system.	-	-	-	-
PC4. Students shall demonstrate the ability to install and configure various input/output devices and peripheral components.	-	-	-	-
<i>Identification and Configuration of Motherboards, Memory, and Storage Devices</i>	-	-	-	-
PC5. Students shall identify different types of motherboards, memory (RAM, ROM), and storage devices (HDD, SSD) and explain their functions within the computer system.	-	-	-	-
PC6. Students shall demonstrate the installation and configuration of motherboards, memory modules, and storage devices.	-	-	-	-
<i>Identification and Use of Computer Ports and Connectors</i>	-	-	-	-
PC7. Students shall identify different types of computer ports (USB, HDMI, Ethernet) and connectors and explain their purposes.	-	-	-	-

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Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
PC8. Students shall demonstrate the proper connection of peripherals and devices using various ports and connectors, ensuring correct setup and functionality.	-	-	-	-
NOS Total	50	22	-	-

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National Occupational Standards (NOS) Parameters

NOS Code	NIE/ELE/N0918
NOS Name	Identification of Computer Hardware Components and Peripherals devices
Sector	Electronics
Sub-Sector	
Occupation	Electronics Assembly
NSQF Level	3
Credits	2
Version	1.0
Last Reviewed Date	25/07/2024
Next Review Date	25/07/2027
NSQC Clearance Date	25/07/2024

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NIE/ELE/N0920: Installation of Computer Hardware Components and setting up of peripherals devices

Description

The NOS focuses on the installation of computer hardware components and the setup of peripheral devices. Upon completion of this module, students will be proficient in installing all necessary components within the CPU cabinet, including the case, fan, power supply, heat sink, CPU, motherboard, and various storage devices such as RAM, hard disk drives, and SSDs. Additionally, students will learn to install and connect peripherals like printers and scanners to the computer system and execute the corresponding setup programs. The module emphasizes the importance of following safety procedures while handling and installing equipment to ensure a secure working environment. Through a combination of theoretical knowledge and practical experience, students will be well-equipped to manage hardware installation and setup tasks effectively in real-world scenarios.

Scope

The scope covers the following :

- This NOS covers the essential training for installing computer hardware components and configuring peripheral devices. Students will gain hands-on experience in assembling internal CPU components, including the motherboard and storage devices. They will also learn to set up and connect peripherals like printers and scanners. Emphasizing safety procedures, this NOS equips students to handle installations securely and effectively in professional settings.

Elements and Performance Criteria

Installation of CPU Components

To be competent, the user/individual on the job must be able to:

- PC1.** Correctly install all components within the CPU cabinet, including the case, fan, power supply, heat sink, CPU, and motherboard.

Installation of Storage Devices

To be competent, the user/individual on the job must be able to:

- PC2.** Install different types of storage devices, including RAM, hard disk drives, and SSDs, according to specifications.
- PC3.** Ensure all storage devices are properly connected to the motherboard and power supply.

Setup of Peripheral Devices

To be competent, the user/individual on the job must be able to:

- PC4.** Successfully install and connect peripheral devices such as printers and scanners to the computer system.
- PC5.** Execute the necessary setup programs for each peripheral device to ensure functionality.

Adherence to Safety Procedures

To be competent, the user/individual on the job must be able to:

- PC6.** Adhere to established safety procedures while handling and installing computer equipment.

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PC7. Demonstrate safe handling techniques to prevent damage to components and ensure personal safety.

Knowledge and Understanding (KU)

The individual on the job needs to know and understand:

- KU1.** Understand the various components of a computer system, including the CPU, motherboard, storage devices (RAM, hard disk drives, SSDs), and peripheral devices (printers, scanners).
- KU2.** Knowledge of proper installation techniques for hardware components within the CPU cabinet, including mounting and securing components.
- KU3.** Familiarity with the connection methods for different peripheral devices and the corresponding software setup required for their operation.
- KU4.** Awareness of safety procedures and best practices for handling and installing computer hardware to minimize risks of damage and ensure a safe working environment.

Generic Skills (GS)

User/individual on the job needs to know how to:

- GS1.** Ability to effectively install and configure various computer hardware components and peripherals, demonstrating technical competence in hardware management.
- GS2.** Skill in troubleshooting installation issues and resolving connectivity problems between hardware components and peripheral devices.
- GS3.** Capability to follow detailed installation procedures and safety protocols meticulously, ensuring accuracy and compliance during hardware setup.
- GS4.** Ability to communicate effectively with team members and clients regarding installation processes, safety practices, and equipment functionality.

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Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
<i>Installation of CPU Components</i>	50	23	-	-
PC1. Correctly install all components within the CPU cabinet, including the case, fan, power supply, heat sink, CPU, and motherboard.	-	-	-	-
<i>Installation of Storage Devices</i>	-	-	-	-
PC2. Install different types of storage devices, including RAM, hard disk drives, and SSDs, according to specifications.	-	-	-	-
PC3. Ensure all storage devices are properly connected to the motherboard and power supply.	-	-	-	-
<i>Setup of Peripheral Devices</i>	-	-	-	-
PC4. Successfully install and connect peripheral devices such as printers and scanners to the computer system.	-	-	-	-
PC5. Execute the necessary setup programs for each peripheral device to ensure functionality.	-	-	-	-
<i>Adherence to Safety Procedures</i>	-	-	-	-
PC6. Adhere to established safety procedures while handling and installing computer equipment.	-	-	-	-
PC7. Demonstrate safe handling techniques to prevent damage to components and ensure personal safety.	-	-	-	-
NOS Total	50	23	-	-

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National Occupational Standards (NOS) Parameters

NOS Code	NIE/ELE/N0920
NOS Name	Installation of Computer Hardware Components and setting up of peripherals devices
Sector	Electronics
Sub-Sector	
Occupation	Electronics Assembly
NSQF Level	3
Credits	2
Version	1.0
Last Reviewed Date	25/07/2024
Next Review Date	25/07/2027
NSQC Clearance Date	25/07/2024

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NIE/ELE/N0919: Installation and Configuration of Computer Software

Description

The NOS focuses on the installation and configuration of computer software, equipping students with essential skills for effective software management. Upon completion of this module, students will understand the booting processes in both DOS and Windows environments and the functionalities of BIOS utilities, including its setup. They will learn to install operating systems and relevant application software while also gaining the ability to reset BIOS and system passwords. This module combines theoretical concepts with practical experience, ensuring students are well-prepared to handle software installation and configuration tasks in various computing environments.

Scope

The scope covers the following :

- The NOS encompasses the essential processes involved in the installation and configuration of computer software, including operating systems and application software. It covers understanding the booting processes in DOS and Windows environments, as well as BIOS utility functions and setup procedures. Students will gain hands-on experience in partitioning hard drives, installing various operating systems, and managing software configurations. Additionally, the module addresses security aspects such as resetting BIOS and system passwords. This comprehensive approach prepares students for real-world scenarios in software management and troubleshooting, ensuring they can effectively support and maintain computer systems.

Elements and Performance Criteria

Boot Processes and BIOS Configuration

To be competent, the user/individual on the job must be able to:

- PC1.** Describe the booting process in DOS and Windows environments.
- PC2.** Access and navigate the BIOS setup utility, configuring boot options and settings effectively.

Disk Management and Operating System Installation

To be competent, the user/individual on the job must be able to:

- PC3.** Format hard disks and partition drives according to specified requirements.
- PC4.** Execute the installation of various operating systems, including configuring dual-boot setups as necessary

Application Software Installation and Configuration

To be competent, the user/individual on the job must be able to:

- PC5.** Identify various types of application software relevant to user needs and install them correctly.
- PC6.** Install and configure antivirus software to provide optimal protection against threats.

Command Prompt Execution and Security Management

To be competent, the user/individual on the job must be able to:

- PC7.** Execute various Windows command prompt commands effectively to manage software and system configurations.
- PC8.** Reset the BIOS password securely while ensuring compliance with security protocols.

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Knowledge and Understanding (KU)

The individual on the job needs to know and understand:

- KU1.** Understand the booting sequence in DOS and Windows environments and the critical role of BIOS utility functions in system settings, such as boot order.
- KU2.** Comprehend the fundamentals of disk formatting, partitioning, and the necessary steps for installing various operating systems, including compatibility checks.
- KU3.** Recognize different types of application software and their functions, along with the procedures for installing and configuring antivirus software to safeguard systems.
- KU4.** Familiarize with essential Windows command prompt commands for system management and understand security practices, including resetting BIOS passwords and managing system access.

Generic Skills (GS)

User/individual on the job needs to know how to:

- GS1.** Develop the ability to navigate and utilize BIOS settings, execute boot configurations, and effectively use command prompt commands for system management.
- GS2.** Enhance troubleshooting capabilities to identify and resolve issues related to operating system installations, software configurations, and hardware compatibility.
- GS3.** Cultivate meticulousness in disk management tasks such as formatting and partitioning drives, ensuring accuracy during the installation of operating systems and applications.
- GS4.** Promote understanding of security protocols, including the importance of resetting BIOS passwords and configuring antivirus software to protect computer systems from threats.

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Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
<i>Boot Processes and BIOS Configuration</i>	60	27	-1	-
PC1. Describe the booting process in DOS and Windows environments.	-	-	-	-
PC2. Access and navigate the BIOS setup utility, configuring boot options and settings effectively.	-	-	-	-
<i>Disk Management and Operating System Installation</i>	-	-	-	-
PC3. Format hard disks and partition drives according to specified requirements.	-	-	-	-
PC4. Execute the installation of various operating systems, including configuring dual-boot setups as necessary	-	-	-	-
<i>Application Software Installation and Configuration</i>	-	-	-	-
PC5. Identify various types of application software relevant to user needs and install them correctly.	-	-	-	-
PC6. Install and configure antivirus software to provide optimal protection against threats.	-	-	-	-
<i>Command Prompt Execution and Security Management</i>	-	-	-	-
PC7. Execute various Windows command prompt commands effectively to manage software and system configurations.	-	-	-	-
PC8. Reset the BIOS password securely while ensuring compliance with security protocols.	-	-	-	-
NOS Total	60	27	-	-

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National Occupational Standards (NOS) Parameters

NOS Code	NIE/ELE/N0919
NOS Name	Installation and Configuration of Computer Software
Sector	Electronics
Sub-Sector	
Occupation	Electronics Assembly
NSQF Level	3
Credits	3
Version	1.0
Last Reviewed Date	25/07/2024
Next Review Date	25/07/2027
NSQC Clearance Date	25/07/2024

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NIE/ELE/N0917: Troubleshooting and Maintenance of Computer System

Description

The NOS concentrates on the troubleshooting and maintenance of computer systems. Upon completion of this module, students will acquire a comprehensive understanding of troubleshooting methodologies and the significance of data backup protocols. They will gain proficiency in executing system utility tools and diagnosing issues related to operating systems, audio components, USB interfaces, and LAN connectivity. Furthermore, students will develop the expertise to identify various computer components, perform installation and configuration of operating systems and drivers, and effectively resolve technical issues as they arise. The module also underscores the necessity of cybersecurity by configuring and utilizing antivirus software to protect against malware and other threats. Through a combination of theoretical knowledge and hands-on experience, students will be equipped to maintain and support computer systems in professional environments.

Scope

The scope covers the following :

- The NOS focuses on the essential skills and knowledge required for troubleshooting and maintaining computer systems. It encompasses a wide range of topics, including preventive and breakdown maintenance, system diagnostics, and the execution of utility tools. Students will learn to identify and troubleshoot various hardware and software issues, take appropriate backup measures, and configure operating systems and drivers effectively. Additionally, the module emphasizes the importance of cybersecurity, guiding students in configuring and running antivirus software to protect against malware and viruses. This comprehensive scope ensures that students are well-prepared to handle real-world technical challenges in computer system maintenance and support.

Elements and Performance Criteria

Understanding Troubleshooting Methods and Backup Procedures

To be competent, the user/individual on the job must be able to:

PC1. Demonstrate knowledge of various troubleshooting methodologies.

PC2. Identify and apply different backup techniques based on system requirements.

Execution of System Utility and Diagnostic Tools

To be competent, the user/individual on the job must be able to:

PC3. Effectively execute system utility tools to diagnose and enhance system performance.

PC4. Analyze the results from diagnostic tools to identify potential system issues.

Recognition and Troubleshooting of Computer Components

To be competent, the user/individual on the job must be able to:

PC5. Identify various hardware components and their functions within a computer system.

PC6. Troubleshoot and resolve issues related to the operating system and its drivers.

Configuration and Management of Antivirus Software

To be competent, the user/individual on the job must be able to:

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- PC7.** Configure antivirus software settings to provide optimal protection against malware and viruses.
- PC8.** Execute scans and updates of antivirus software to ensure system security.

Knowledge and Understanding (KU)

The individual on the job needs to know and understand:

- KU1.** Understand how to diagnose and resolve issues with sound, USB, LAN, and other components, including identifying faults in power supplies and expansion cards.
- KU2.** Gain knowledge of backing up, restoring drivers, and recovering lost or corrupted data effectively.
- KU3.** Learn to utilize system utility software and diagnostic tools for maintaining, optimizing, and troubleshooting system performance.
- KU4.** Understand backup methods for data security and learn how to configure antivirus software to protect systems from malware and viruses.

Generic Skills (GS)

User/individual on the job needs to know how to:

- GS1.** Ability to troubleshoot hardware issues, including sound, USB, and LAN problems, and accurately diagnose faults in power supplies and expansion cards.
- GS2.** Skilled in installing and configuring drivers, executing backup and recovery procedures, and running diagnostic tools for system maintenance.
- GS3.** Capable of performing data backup, restoration, and fine-tuning system settings to ensure smooth operations and data integrity.
- GS4.** Competent in running antivirus programs, identifying potential security threats, and protecting systems from malware, viruses, and other vulnerabilities.

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Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
<i>Understanding Troubleshooting Methods and Backup Procedures</i>	40	18	-	-
PC1. Demonstrate knowledge of various troubleshooting methodologies.	-	-	-	-
PC2. Identify and apply different backup techniques based on system requirements.	-	-	-	-
<i>Execution of System Utility and Diagnostic Tools</i>	-	-	-	-
PC3. Effectively execute system utility tools to diagnose and enhance system performance.	-	-	-	-
PC4. Analyze the results from diagnostic tools to identify potential system issues.	-	-	-	-
<i>Recognition and Troubleshooting of Computer Components</i>	-	-	-	-
PC5. Identify various hardware components and their functions within a computer system.	-	-	-	-
PC6. Troubleshoot and resolve issues related to the operating system and its drivers.	-	-	-	-
<i>Configuration and Management of Antivirus Software</i>	-	-	-	-
PC7. Configure antivirus software settings to provide optimal protection against malware and viruses.	-	-	-	-
PC8. Execute scans and updates of antivirus software to ensure system security.	-	-	-	-
NOS Total	40	18	-	-

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National Occupational Standards (NOS) Parameters

NOS Code	NIE/ELE/N0917
NOS Name	Troubleshooting and Maintenance of Computer System
Sector	Electronics
Sub-Sector	
Occupation	Electronics Assembly
NSQF Level	3
Credits	2
Version	1.0
Last Reviewed Date	25/07/2024
Next Review Date	25/07/2027
NSQC Clearance Date	25/07/2024

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NIE/ELE/N0921: Comprehensive Computer System Setup, Installation, and Maintenance Project

Description

This National Occupational Standard (NOS) outlines the essential skills and knowledge required for successfully setting up, installing, and maintaining computer systems. It emphasizes proficiency in identifying computer components, assembling hardware, installing software, and troubleshooting system issues. Key competencies include the ability to recognize various computer peripherals, configure BIOS settings, install operating systems, and execute system utility tools. Effective handling of hardware components, adherence to safety protocols, and diagnosing system errors are also critical components.

Scope

The scope covers the following :

- The scope of the National Occupational Standard (NOS) for Computer System Setup, Installation, and Maintenance encompasses the complete process, from identifying components to troubleshooting and system optimization. It includes proficiency in recognizing various hardware and peripherals, assembling and configuring computer systems, and installing operating systems and software. Emphasis is placed on following safety procedures, executing system utility tools, and ensuring system security through antivirus configuration. Effective problem-solving skills and adherence to industry best practices are essential for maintaining system reliability and performance.

Elements and Performance Criteria

Computer System Identification, Installation, and Maintenance

To be competent, the user/individual on the job must be able to:

- PC1.** Identify different types of computer systems, hardware components, peripherals, and their functions
- PC2.** Install and configure computer hardware components, peripheral devices, and necessary software.
- PC3.** Troubleshoot system errors, perform data backup, and optimize system performance using utility tools.
- PC4.** Implement security measures, including BIOS configuration, system password management, and antivirus installation.

Knowledge and Understanding (KU)

The individual on the job needs to know and understand:

- KU1.** Identify different types of computer systems, their working principles, and differentiate between hardware and software components.
- KU2.** Understand the installation and configuration processes for computer hardware, peripheral devices, and software applications.

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- KU3.** Learn troubleshooting techniques, system utility tools, and backup methods to maintain system performance and reliability.
- KU4.** Gain knowledge of BIOS settings, system security measures, and the implementation of antivirus software for system protection.

Generic Skills (GS)

User/individual on the job needs to know how to:

- GS1.** Develop expertise in identifying and assembling various computer hardware components, including motherboards, memory, storage devices, and peripherals, ensuring compatibility and proper installation.
- GS2.** Cultivate strong analytical skills to diagnose and troubleshoot hardware and software issues, including system errors, connectivity problems, and performance bottlenecks, using appropriate tools and techniques.
- GS3.** Enhance the ability to effectively communicate technical concepts related to computer hardware, software installation, and troubleshooting. Prepare clear documentation for system configurations, installation procedures, and maintenance logs.
- GS4.** Apply problem-solving techniques to configure BIOS settings, install operating systems and applications, and implement system security measures, ensuring optimal system functionality and protection against threats.

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Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
<i>Computer System Identification, Installation, and Maintenance</i>	-	-	-	-
PC1. Identify different types of computer systems, hardware components, peripherals, and their functions	-	-	-	-
PC2. Install and configure computer hardware components, peripheral devices, and necessary software.	-	-	-	-
PC3. Troubleshoot system errors, perform data backup, and optimize system performance using utility tools.	-	-	-	-
PC4. Implement security measures, including BIOS configuration, system password management, and antivirus installation.	-	-	-	-
NOS Total	-	-	30	-

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National Occupational Standards (NOS) Parameters

NOS Code	NIE/ELE/N0921
NOS Name	Comprehensive Computer System Setup, Installation, and Maintenance Project
Sector	Electronics
Sub-Sector	
Occupation	Electronics Assembly
NSQF Level	3
Credits	1
Version	1.0
Last Reviewed Date	25/07/2024
Next Review Date	25/07/2027
NSQC Clearance Date	25/07/2024

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DGT/VSQ/N0101: Employability Skills (30 Hours)

Description

This unit is about employability skills, Constitutional values, becoming a professional in the 21st Century, digital, financial, and legal literacy, diversity and Inclusion, English and communication skills, customer service, entrepreneurship, and apprenticeship, getting ready for jobs and career development.

Scope

The scope covers the following :

- Introduction to Employability Skills
- Constitutional values - Citizenship
- Becoming a Professional in the 21st Century
- Basic English Skills
- Communication Skills
- Diversity & Inclusion
- Financial and Legal Literacy
- Essential Digital Skills
- Entrepreneurship
- Customer Service
- Getting ready for Apprenticeship & Jobs

Elements and Performance Criteria

Introduction to Employability Skills

To be competent, the user/individual on the job must be able to:

PC1. understand the significance of employability skills in meeting the job requirements

Constitutional values – Citizenship

To be competent, the user/individual on the job must be able to:

PC2. identify constitutional values, civic rights, duties, personal values and ethics and environmentally sustainable practices

Becoming a Professional in the 21st Century

To be competent, the user/individual on the job must be able to:

PC3. explain 21st Century Skills such as Self-Awareness, Behavior Skills, Positive attitude, self-motivation, problem-solving, creative thinking, time management, social and cultural awareness, emotional awareness, continuous learning mindset etc.

Basic English Skills

To be competent, the user/individual on the job must be able to:

PC4. speak with others using some basic English phrases or sentences

Communication Skills

To be competent, the user/individual on the job must be able to:

PC5. follow good manners while communicating with others

PC6. work with others in a team

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Diversity & Inclusion

To be competent, the user/individual on the job must be able to:

PC7. communicate and behave appropriately with all genders and PwD

PC8. report any issues related to sexual harassment

Financial and Legal Literacy

To be competent, the user/individual on the job must be able to:

PC9. use various financial products and services safely and securely

PC10. calculate income, expenses, savings etc.

PC11. approach the concerned authorities for any exploitation as per legal rights and laws

Essential Digital Skills

To be competent, the user/individual on the job must be able to:

PC12. operate digital devices and use its features and applications securely and safely

PC13. use internet and social media platforms securely and safely

Entrepreneurship

To be competent, the user/individual on the job must be able to:

PC14. identify and assess opportunities for potential business

PC15. identify sources for arranging money and associated financial and legal challenges

Customer Service

To be competent, the user/individual on the job must be able to:

PC16. identify different types of customers

PC17. identify customer needs and address them appropriately

PC18. follow appropriate hygiene and grooming standards

Getting ready for apprenticeship & Jobs

To be competent, the user/individual on the job must be able to:

PC19. create a basic biodata

PC20. search for suitable jobs and apply

PC21. identify and register apprenticeship opportunities as per requirement

Knowledge and Understanding (KU)

The individual on the job needs to know and understand:

KU1. need for employability skills

KU2. various constitutional and personal values

KU3. different environmentally sustainable practices and their importance

KU4. Twenty first (21st) century skills and their importance

KU5. how to use basic spoken English language

KU6. Do and dont of effective communication

KU7. inclusivity and its importance

KU8. different types of disabilities and appropriate communication and behaviour towards PwD

KU9. different types of financial products and services

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- KU10.** how to compute income and expenses
- KU11.** importance of maintaining safety and security in financial transactions
- KU12.** different legal rights and laws
- KU13.** how to operate digital devices and applications safely and securely
- KU14.** ways to identify business opportunities
- KU15.** types of customers and their needs
- KU16.** how to apply for a job and prepare for an interview
- KU17.** apprenticeship scheme and the process of registering on apprenticeship portal

Generic Skills (GS)

User/individual on the job needs to know how to:

- GS1.** communicate effectively using appropriate language
- GS2.** behave politely and appropriately with all
- GS3.** perform basic calculations
- GS4.** solve problems effectively
- GS5.** be careful and attentive at work
- GS6.** use time effectively
- GS7.** maintain hygiene and sanitisation to avoid infection

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Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
<i>Introduction to Employability Skills</i>	1	1	-	-
PC1. understand the significance of employability skills in meeting the job requirements	-	-	-	-
<i>Constitutional values – Citizenship</i>	1	1	-	-
PC2. identify constitutional values, civic rights, duties, personal values and ethics and environmentally sustainable practices	-	-	-	-
<i>Becoming a Professional in the 21st Century</i>	1	3	-	-
PC3. explain 21st Century Skills such as Self-Awareness, Behavior Skills, Positive attitude, self-motivation, problem-solving, creative thinking, time management, social and cultural awareness, emotional awareness, continuous learning mindset etc.	-	-	-	-
<i>Basic English Skills</i>	2	3	-	-
PC4. speak with others using some basic English phrases or sentences	-	-	-	-
<i>Communication Skills</i>	1	1	-	-
PC5. follow good manners while communicating with others	-	-	-	-
PC6. work with others in a team	-	-	-	-
<i>Diversity & Inclusion</i>	1	1	-	-
PC7. communicate and behave appropriately with all genders and PwD	-	-	-	-
PC8. report any issues related to sexual harassment	-	-	-	-
<i>Financial and Legal Literacy</i>	3	4	-	-
PC9. use various financial products and services safely and securely	-	-	-	-

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Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
PC10. calculate income, expenses, savings etc.	-	-	-	-
PC11. approach the concerned authorities for any exploitation as per legal rights and laws	-	-	-	-
<i>Essential Digital Skills</i>	4	6	-	-
PC12. operate digital devices and use its features and applications securely and safely	-	-	-	-
PC13. use internet and social media platforms securely and safely	-	-	-	-
<i>Entrepreneurship</i>	3	5	-	-
PC14. identify and assess opportunities for potential business	-	-	-	-
PC15. identify sources for arranging money and associated financial and legal challenges	-	-	-	-
<i>Customer Service</i>	2	2	-	-
PC16. identify different types of customers	-	-	-	-
PC17. identify customer needs and address them appropriately	-	-	-	-
PC18. follow appropriate hygiene and grooming standards	-	-	-	-
<i>Getting ready for apprenticeship & Jobs</i>	1	3	-	-
PC19. create a basic biodata	-	-	-	-
PC20. search for suitable jobs and apply	-	-	-	-
PC21. identify and register apprenticeship opportunities as per requirement	-	-	-	-
NOS Total	20	30	-	-

Qualification Pack

National Occupational Standards (NOS) Parameters

NOS Code	DGT/VSQ/N0101
NOS Name	Employability Skills (30 Hours)
Sector	Cross Sectoral
Sub-Sector	Professional Skills
Occupation	Employability
NSQF Level	2
Credits	1
Version	1.0
Last Reviewed Date	12/03/2026
Next Review Date	12/03/2029
NSQC Clearance Date	12/03/2026

Assessment Guidelines and Assessment Weightage

Assessment Guidelines

Assessment of the qualification evaluates candidates to ascertain that they can integrate knowledge, skills and values for carrying out relevant tasks as per the defined learning outcomes and assessment criteria.

The underlying principle of assessment is fairness and transparency. The evidence of the outcomes and assessment criteria. Competence acquired by the candidate can be obtained by conducting Theory (Online), Practical assessment, Internal assessment, Project/Presentation/ Assignment, Major Project. The emphasis is on the practical demonstration of skills & knowledge gained by the candidate through the training. Each OUTCOME is assessed & marked separately. A candidate is required to pass all OUTCOMES individually based on the passing criteria.

About Examination Pattern:

1. The question papers for the theory and practical exams are set by the Examination wing (assessor) of NIELIT HQS.
2. The assessor assigns roll number.
3. The assessor carries out theory online assessments through remote proctoring methodology. Theory

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examination would be conducted online and the paper comprise of MCQ. Conduct of assessment are through trained proctors. Once the test begins, remote proctors have full access to candidate's video feeds and computer screens. Proctors authenticate the candidate based on registration details, pre-test image captured and I- card in possession of the candidate. Proctors can chat with candidates or give warnings to candidates. Proctors can also take screenshots, terminate a specific user's test session, or re-authenticate candidates based on video feeds.

4. An External Examiner/ Observer may be deployed including NIELIT officials for evaluation of Practical examination/ internal assessment / Project/ Presentation/. Major Project (if applicable) would be evaluated preferably by external/ subject expert including NIELIT officials.

5. Pass percentage would be 50% marks in each component.

6. Candidates may apply for re-examination within the validity of registration (only in the assessment component in which the candidate failed).

7. For re-examination prescribed examination fee is required to be paid by the candidate only for the assessment component in which the candidate wants to reappear.

8. There would be no exemption for any paper/module for candidates having similar qualifications or skills.

9. The examination will be conducted in English language only.

Quality assurance activities: A pool of questions is created by a subject matter expert and moderated by other SME. Test rules are set beforehand. Random set of questions which are according to syllabus appears which may differ from candidate to candidate. Confidentiality and impartiality are maintained during all the examination and evaluation processes.

Minimum Aggregate Passing % at QP Level : 50

(Please note: Every Trainee should score a minimum aggregate passing percentage as specified above, to successfully clear the Qualification Pack assessment.)

Assessment Weightage

Compulsory NOS

National Occupational Standards	Theory Marks	Practical Marks	Project Marks	Viva Marks	Total Marks	Weightage
NIE/ELE/N0918.Identification of Computer Hardware Components and Peripherals devices	50	22	0	0	72	20

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National Occupational Standards	Theory Marks	Practical Marks	Project Marks	Viva Marks	Total Marks	Weightage
NIE/ELE/N0920.Installation of Computer Hardware Components and setting up of peripherals devices	50	23	0	0	73	21
NIE/ELE/N0919.Installation and Configuration of Computer Software	60	27	-	-	87	25
NIE/ELE/N0917.Troubleshooting and Maintenance of Computer System	40	18	-	-	58	17
NIE/ELE/N0921.Comprehensive Computer System Setup, Installation, and Maintenance Project	-	-	30	-	30	9
DGT/VSQ/N0101.Employability Skills (30 Hours)	20	30	-	-	50	8
Total	220	120	30	-	370	100

Qualification Pack

Acronyms

NOS	National Occupational Standard(s)
NSQF	National Skills Qualifications Framework
QP	Qualifications Pack
TVET	Technical and Vocational Education and Training
AA	Assessment Agency
AB	Awarding Body
ISCO	International Standard Classification of Occupations
NCO	National Classification of Occupations
NCrF	National Credit Framework
NOS	National Occupational Standard(s)
NQR	National Qualification Register
NSQF	National Skills Qualifications Framework
OJT	On the Job Training
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Qualification Pack

Glossary

Sector	Sector is a conglomeration of different business operations having similar business and interests. It may also be defined as a distinct subset of the economy whose components share similar characteristics and interests.
Sub-sector	Sub-sector is derived from a further breakdown based on the characteristics and interests of its components.
Occupation	Occupation is a set of job roles, which perform similar/ related set of functions in an industry.
Job role	Job role defines a unique set of functions that together form a unique employment opportunity in an organisation.
Occupational Standards (OS)	OS specify the standards of performance an individual must achieve when carrying out a function in the workplace, together with the Knowledge and Understanding (KU) they need to meet that standard consistently. Occupational Standards are applicable both in the Indian and global contexts.
Performance Criteria (PC)	Performance Criteria (PC) are statements that together specify the standard of performance required when carrying out a task.
National Occupational Standards (NOS)	NOS are occupational standards which apply uniquely in the Indian context.
Qualifications Pack (QP)	QP comprises the set of OS, together with the educational, training and other criteria required to perform a job role. A QP is assigned a unique qualifications pack code.
Unit Code	Unit code is a unique identifier for an Occupational Standard, which is denoted by an 'N'
Unit Title	Unit title gives a clear overall statement about what the incumbent should be able to do.
Description	Description gives a short summary of the unit content. This would be helpful to anyone searching on a database to verify that this is the appropriate OS they are looking for.
Scope	Scope is a set of statements specifying the range of variables that an individual may have to deal with in carrying out the function which have a critical impact on quality of performance required.
Knowledge and Understanding (KU)	Knowledge and Understanding (KU) are statements which together specify the technical, generic, professional and organisational specific knowledge that an individual needs in order to perform to the required standard.

Qualification Pack

Organisational Context	Organisational context includes the way the organisation is structured and how it operates, including the extent of operative knowledge managers have of their relevant areas of responsibility.
Technical Knowledge	Technical knowledge is the specific knowledge needed to accomplish specific designated responsibilities.
Core Skills/ Generic Skills (GS)	Core skills or Generic Skills (GS) are a group of skills that are the key to learning and working in today's world. These skills are typically needed in any work environment in today's world. These skills are typically needed in any work environment. In the context of the OS, these include communication related skills that are applicable to most job roles.
Electives	Electives are NOS/set of NOS that are identified by the sector as contributive to specialization in a job role. There may be multiple electives within a QP for each specialized job role. Trainees must select at least one elective for the successful completion of a QP with Electives.
Options	Options are NOS/set of NOS that are identified by the sector as additional skills. There may be multiple options within a QP. It is not mandatory to select any of the options to complete a QP with Options.
National Occupational Standard	NOS define the measurable performance outcomes required from an individual engaged in a particular task. They list down what an individual performing that task should know and also do.
Qualification	A formal outcome of an assessment and validation process which is obtained when a competent body determines that an individual has achieved learning outcomes to given standards.
Qualification File	A Qualification File is a template designed to capture necessary information of a Qualification from the perspective of NSQF compliance. The Qualification File will be normally submitted by the awarding body for the qualification.
Sector	A grouping of professional activities on the basis of their main economic function, product, service or technology.
Long Term Training	Long-term skilling means any vocational training program undertaken for a year and above. https://ncvet.gov.in/sites/default/files/NCVET.pdf
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